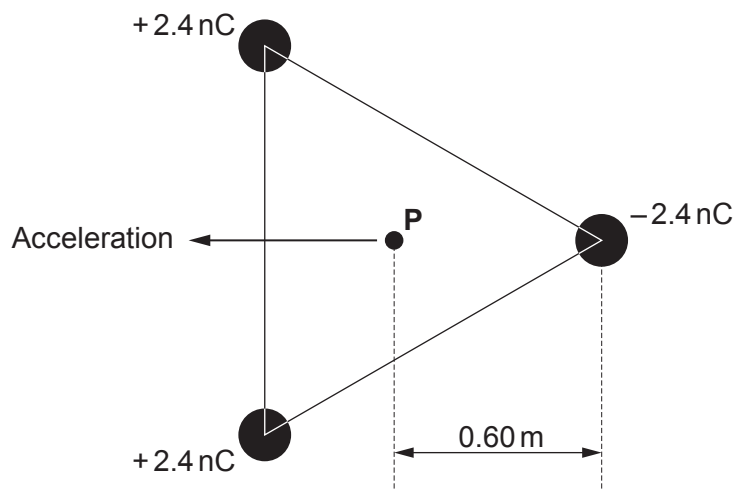


3. Three stationary charges are located at the corners of an **equilateral** triangle and an electron is located at the centre, **P**, of the triangle as shown. The electron is exactly 0.60 m from each charge.



- (a) Draw arrows to show the 3 electric fields at **P** due to the 3 charges. Use these to explain why the electron will accelerate in the direction shown. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



(b) Calculate the initial acceleration of the electron.

[4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) Show **clearly** that the initial potential energy of the electron is approximately $-6 \times 10^{-18} \text{ J}$.

[2]

.....

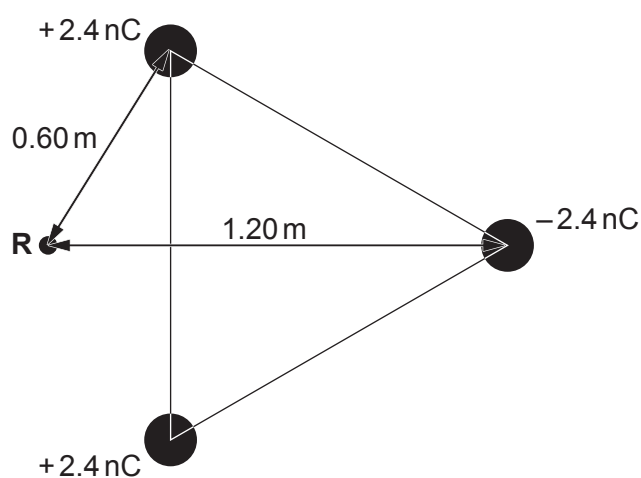
.....

.....

.....



- (d) Determine whether or not the electron (initially at rest) will reach point **R**, shown in the diagram below (both positive charges are 0.60 m from **R**). Explain your reasoning. [4]



.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

